

## INT-332 Practical-5

### Docker-Compose

**Exercise 1:** The following configuration runs an Apache HTTP server.

version: "3"

service:

web:

image: httpd

Modify the file to

1. Map port 8081 on host to port 80 in container.
2. Mount a local folder ./website to /usr/local/apache2/htdocs
3. Set container name as apache\_server

### Answer:

Windows PowerShell

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PS C:\Users\sr302> mkdir apache-compose1

Directory: C:\Users\sr302

| Mode   | LastWriteTime     | Length | Name            |
|--------|-------------------|--------|-----------------|
| d----- | 3/18/2026 8:43 PM |        | apache-compose1 |

PS C:\Users\sr302> cd apache-compose1

PS C:\Users\sr302\apache-compose1> mkdir website

Directory: C:\Users\sr302\apache-compose1

| Mode   | LastWriteTime     | Length | Name    |
|--------|-------------------|--------|---------|
| d----- | 3/18/2026 8:44 PM |        | website |

PS C:\Users\sr302\apache-compose1> notepad index.html

PS C:\Users\sr302\apache-compose1> notepad docker-compose.yml

PS C:\Users\sr302\apache-compose1> dir

Directory: C:\Users\sr302\apache-compose1

| Mode   | LastWriteTime     | Length | Name               |
|--------|-------------------|--------|--------------------|
| d----- | 3/18/2026 8:44 PM |        | website            |
| -a---- | 3/18/2026 8:45 PM | 160    | docker-compose.yml |
| -a---- | 3/18/2026 8:44 PM | 153    | index.html         |

PS C:\Users\sr302\apache-compose1> docker compose up -d

time="2026-03-18T20:45:56+05:30" level=warning msg="No services to build"

[+] up 2/2

✓ Network apache-compose1\_default Created0.1s

✓ Container apache\_serverCreated0.2s

PS C:\Users\sr302\apache-compose1> docker ps

| CONTAINER ID | IMAGE                  | COMMAND                  | CREATED                | STATUS                | PORTS                                           | NAMES         |
|--------------|------------------------|--------------------------|------------------------|-----------------------|-------------------------------------------------|---------------|
| 68b4729e3881 | httpd                  | "httpd-foreground"       | Less than a second ago | Up Less than a second | 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp         | apache_server |
| ec1790ff124b | node-mongo-compose-app | "docker-entrypoint.s..." | 6 days ago             | Up 2 minutes          | 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp     | node_app      |
| 7949411fbc18 | mongo:6                | "docker-entrypoint.s..." | 6 days ago             | Up 2 minutes          | 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp | mongo_db      |
| d6e4baf1ad57 | wordpress:latest       | "docker-entrypoint.s..." | 7 days ago             | Up 2 minutes          | 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp         | wordpress_app |
| 351740cffa5e | mysql:8.0              | "docker-entrypoint.s..." | 7 days ago             | Up 2 minutes          | 3306/tcp, 33060/tcp                             | wordpress_db  |

PS C:\Users\sr302\apache-compose1> ^C

PS C:\Users\sr302\apache-compose1> cd C:\Users\sr302\apache-compose1

>> dir

Directory: C:\Users\sr302\apache-compose1

| Mode   | LastWriteTime     | Length | Name               |
|--------|-------------------|--------|--------------------|
| d----- | 3/18/2026 8:44 PM |        | website            |
| -a---- | 3/18/2026 8:45 PM | 160    | docker-compose.yml |
| -a---- | 3/18/2026 8:44 PM | 153    | index.html         |

PS C:\Users\sr302\apache-compose1>cd website

>> dir

PS C:\Users\sr302\apache-compose1\website> dir

PS C:\Users\sr302\apache-compose1\website>notepad index.html

```
PS C:\Users\sr302\apache-compose1\website> cd ..
```

```
PS C:\Users\sr302\apache-compose1> docker compose restart
```

```
[+] restart 0/1
```

```
- Container apache_server Restarting1.7s
```

```
PS C:\Users\sr302\apache-compose1>
```

## Output:



## Apache Server Running Successfully

**Exercise 2:** Create a docker-compose.yml file to deploy the following application Requirements.

1. Service1: Nginx web server

Image: nginx

Port Mapping : 8085:80

2. Service2: Redis Database

Image: redis

3. Both service must:

Run on a custom network called appnetwork

4. The Nginx container name should be nginx\_server

## Answer:

Windows PowerShell

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```
PS C:\Users\sr302> mkdir nginx-redis-compose
```

Directory: C:\Users\sr302

| Mode   | LastWriteTime     | Length | Name                |
|--------|-------------------|--------|---------------------|
| d----- | 3/18/2026 9:00 PM |        | nginx-redis-compose |

```
PS C:\Users\sr302> cd nginx-redis-compose
```

```
PS C:\Users\sr302\nginx-redis-compose> notepad docker-compose.yml
```

```
PS C:\Users\sr302\nginx-redis-compose> dir
```

Directory: C:\Users\sr302\nginx-redis-compose

| Mode   | LastWriteTime     | Length | Name               |
|--------|-------------------|--------|--------------------|
| -a---- | 3/18/2026 9:01 PM | 232    | docker-compose.yml |

```
PS C:\Users\sr302\nginx-redis-compose>docker compose up -d
```

```
[+] up 10/10
```

```
[+] up 13/13is Pulled47.9s
```

```
✓ Image redisPulled47.9s
```

```
✓ Network nginx-redis-compose_appnetwork Created0.1s
```

```
✓ Container nginx-redis-compose-redis-1 Created0.5s
```

```
✓ Container nginx_serverCreated0.5s
```

```
PS C:\Users\sr302\nginx-redis-compose> docker ps
```

| CONTAINER ID | IMAGE                  | COMMAND                 | CREATED        | STATUS        | PORTS                                           | NAMES                       |
|--------------|------------------------|-------------------------|----------------|---------------|-------------------------------------------------|-----------------------------|
| 985163478fa9 | nginx                  | "/docker-entypoint...." | 3 minutes ago  | Up 4 minutes  | 0.0.0.0:8085->80/tcp, [::]:8085->80/tcp         | nginx_server                |
| 9918185d4fbc | redis                  | "docker-entypoint.s..." | 3 minutes ago  | Up 4 minutes  | 6379/tcp                                        | nginx-redis-compose-redis-1 |
| 68b4729e3881 | httpd                  | "httpd-foreground"      | 20 minutes ago | Up 16 minutes | 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp         | apache_server               |
| ec1790ff124b | node-mongo-compose-app | "docker-entypoint.s..." | 6 days ago     | Up 23 minutes | 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp     | node_app                    |
| 7949411fbc18 | mongo:6                | "docker-entypoint.s..." | 6 days ago     | Up 23 minutes | 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp | mongo_db                    |
| d6e4baf1ad57 | wordpress:latest       | "docker-entypoint.s..." | 7 days ago     | Up 23 minutes | 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp         | wordpress_app               |
| 351740cffa5e | mysql:8.0              | "docker-entypoint.s..." | 7 days ago     | Up 23 minutes | 3306/tcp, 33060/tcp                             | wordpress_db                |

```
PS C:\Users\sr302\nginx-redis-compose> docker network ls
```

| NETWORK ID   | NAME                     | DRIVER | SCOPE |
|--------------|--------------------------|--------|-------|
| 95b9ed4d3f18 | apache-compose1_default  | bridge | local |
| d6a489bbbc23 | bridge                   | bridge | local |
| e31c0768a465 | docker-auto-work_default | bridge | local |

```
e6c68302e88c host          host local
f58ce8443505 nginx-redis-compose_appnetwork bridge local
af1debce232a node-mongo-compose_default bridge local
b99385c1d1bb none          null local
1bde8c9b9a37 wordpress-compose_default bridge local
PS C:\Users\sr302\nginx-redis-compose>
```

## Output:

